



University
of Glasgow

Power Electronics

DC-AC Converters (Inverters)

逆变器

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Please read **Chapter 8** in the textbook

Schemes of Inverters to study

Phase Output	Bridge	Switching Schemes	• Operation in different switching schemes • Output Waveform • Harmonics • Practical Circuit • Avoiding shoot through
Single Phase	Half	Square Wave	
		Pules Width Modulation	
	Full	Square Wave	
		Pules Width Modulation (Bipolar & Unipolar)	
Three Phase	3-Leg	Square Wave	
		Pules Width Modulation	

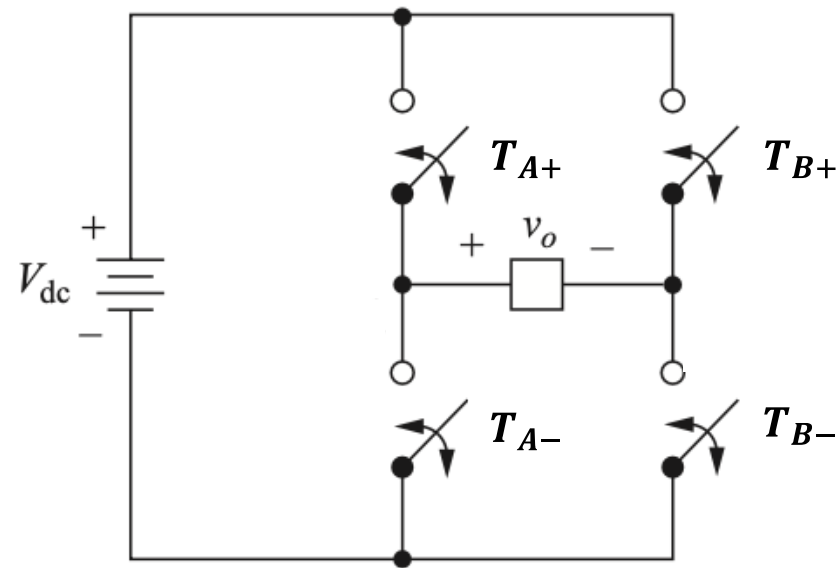
Half Bridge Studied in Last Lecture

Schemes of Inverters to study

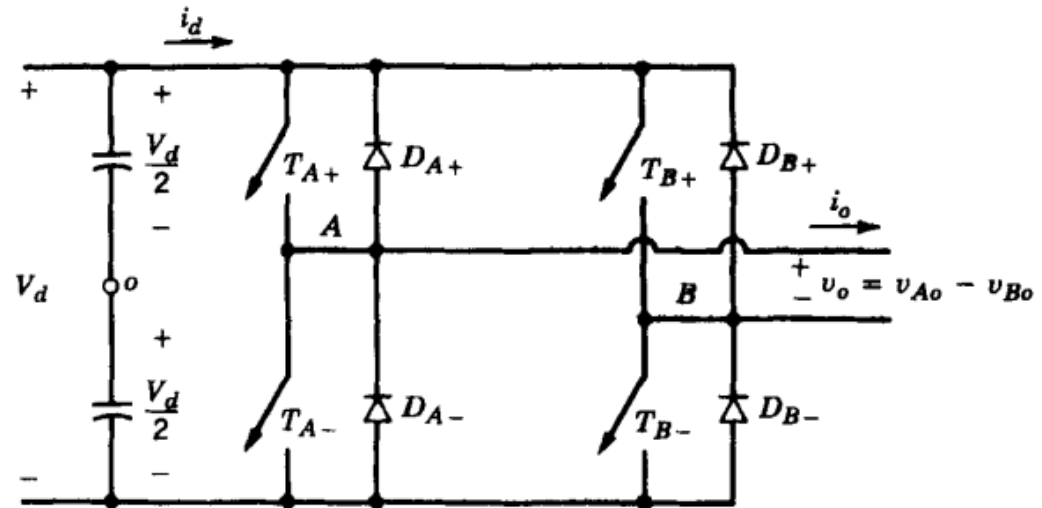
Phase Output	Bridge	Switching Schemes		• Ideal Circuit Diagram • Operation • Output Waveform • Harmonics • Practical Circuit Diagram • Avoiding shoot through
Single Phase	Half	Square Wave		
		Pules Width Modulation		
	Full	Square Wave		
		Pules Width Modulation		
		Bipolar	Unipolar	
Three Phase	3-Leg	Square Wave		
		Pules Width Modulation		

Inverter Types

- Full Bridge



Full Bridge Inverter (Ideal)

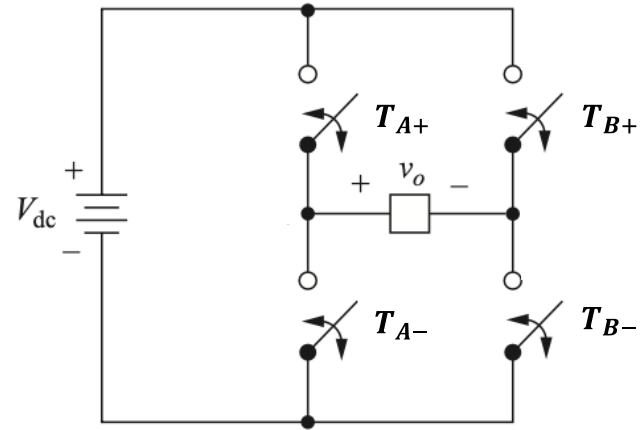
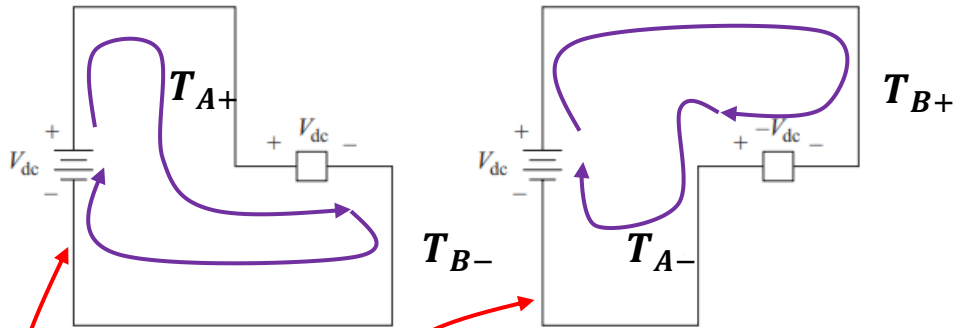


Full Bridge Inverter with Flyback Diodes

Figures used in the literature for Full Bridge Inverter

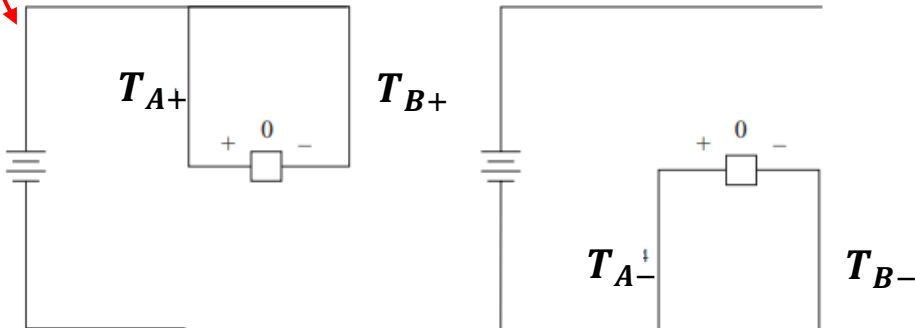
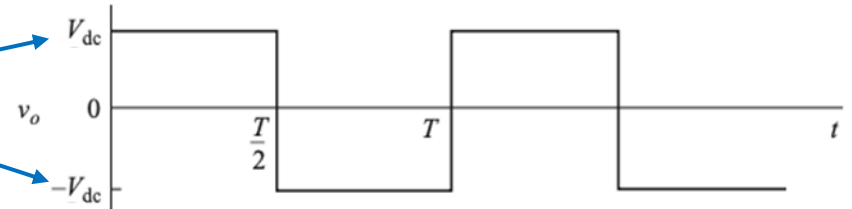
Operation

- Square Wave Switching



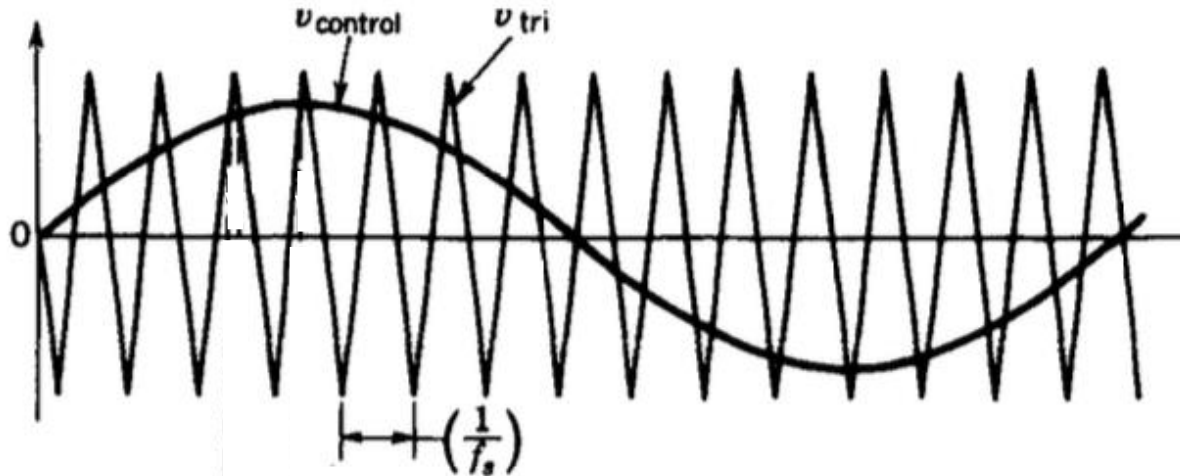
Full Bridge Inverter (Ideal)

Switches Closed	Output Voltage v_o
T_{A+} and T_{B-}	$+V_{dc}$
T_{B+} and T_{A-}	$-V_{dc}$
T_{A+} and T_{B+}	0
T_{A-} and T_{B-}	0



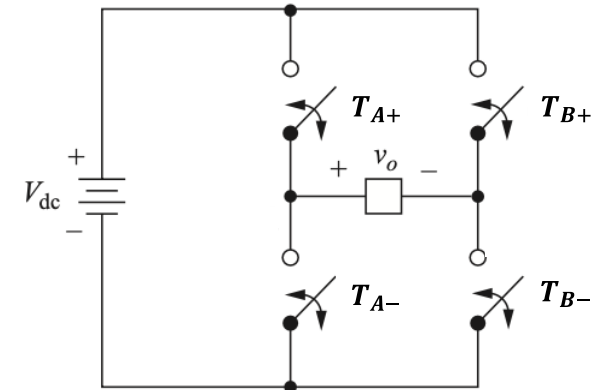
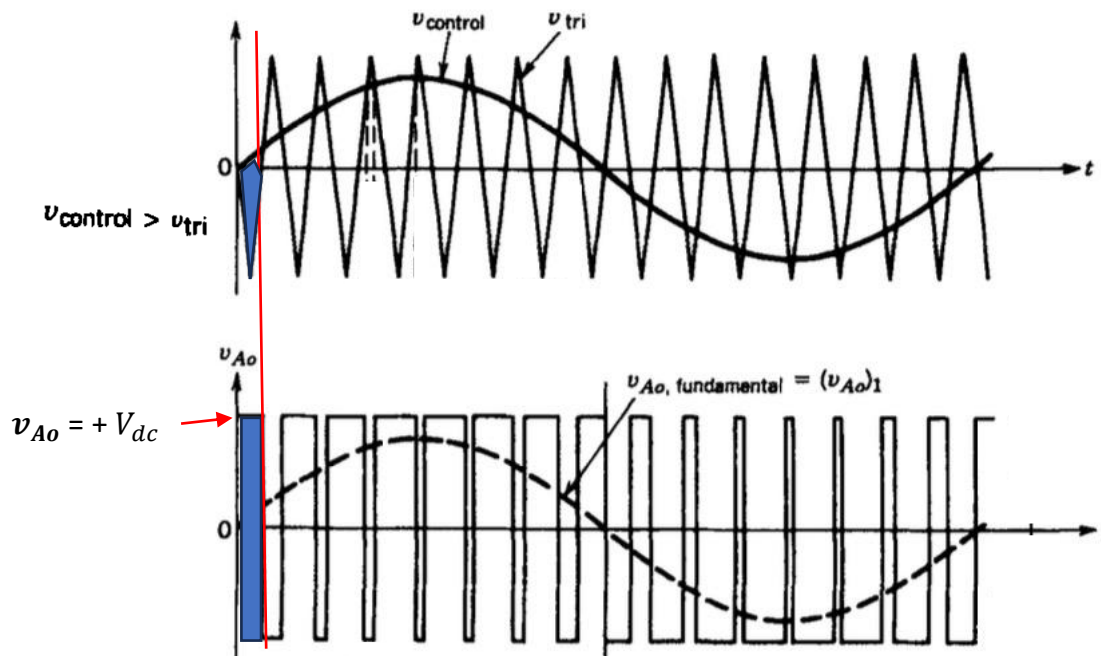
Operation

- Pulse Width Modulation (PWM) Switching - Bipolar
 - Is a type of switching technique.
 - Use two signals called
 - v_{tri} (Triangular Signal)
 - Has frequency f_s (called switching frequency)
 - $v_{control}$ (Control Signal)
 - Has frequency f_1 (desired fundamental frequency)

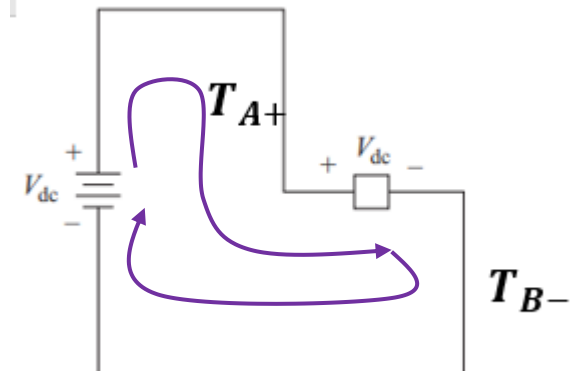


Operation

- Pulse Width Modulation (PWM) Switching - Bipolar
 - v_{tri} (Triangular Signal) is compared with $v_{control}$ (Control Signal)
 - Based on comparison result different switches are made closed (ON) and open (OFF)
 - Output voltage swings between positive and negative voltages

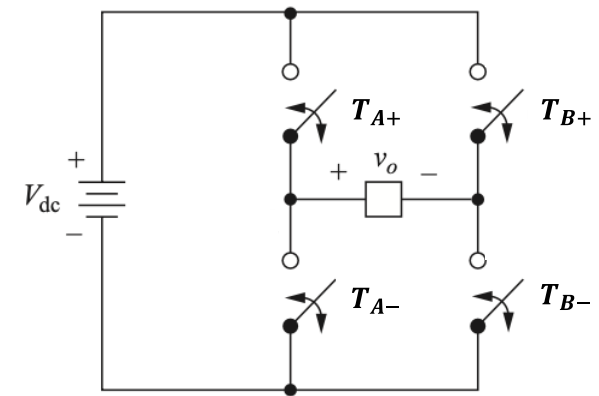
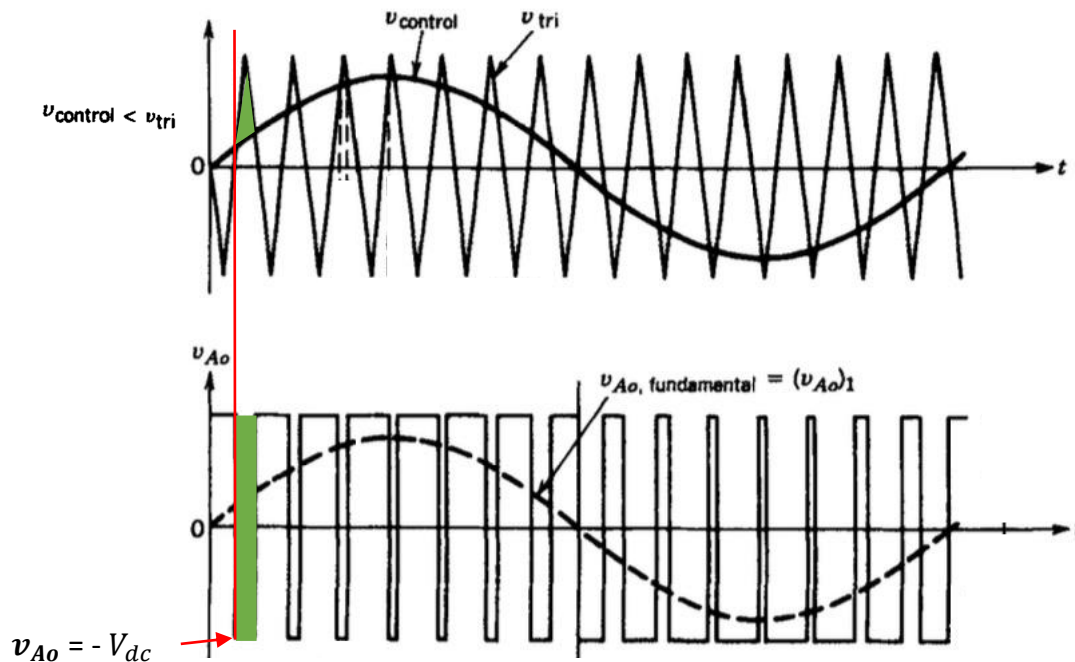


T_{A+} and T_{B-} Closed

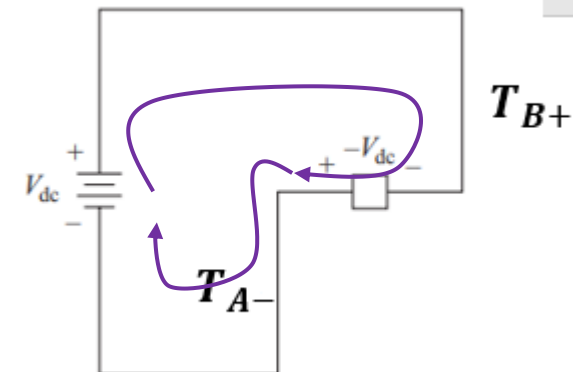


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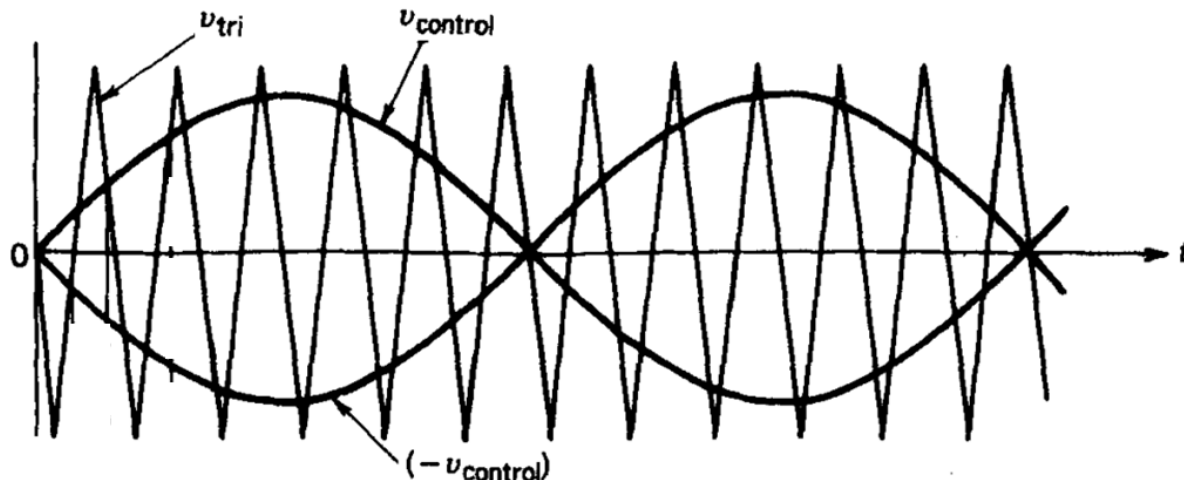


T_{B+} and T_{A-} Closed



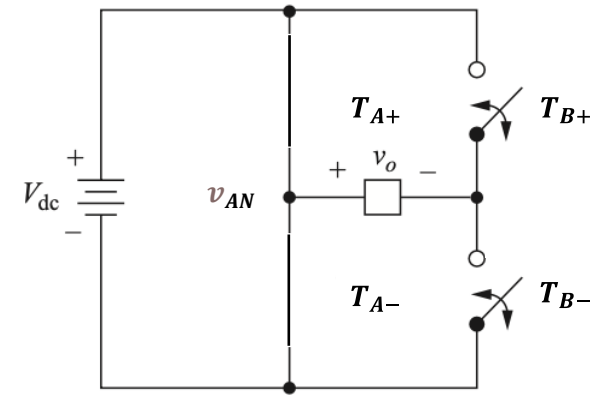
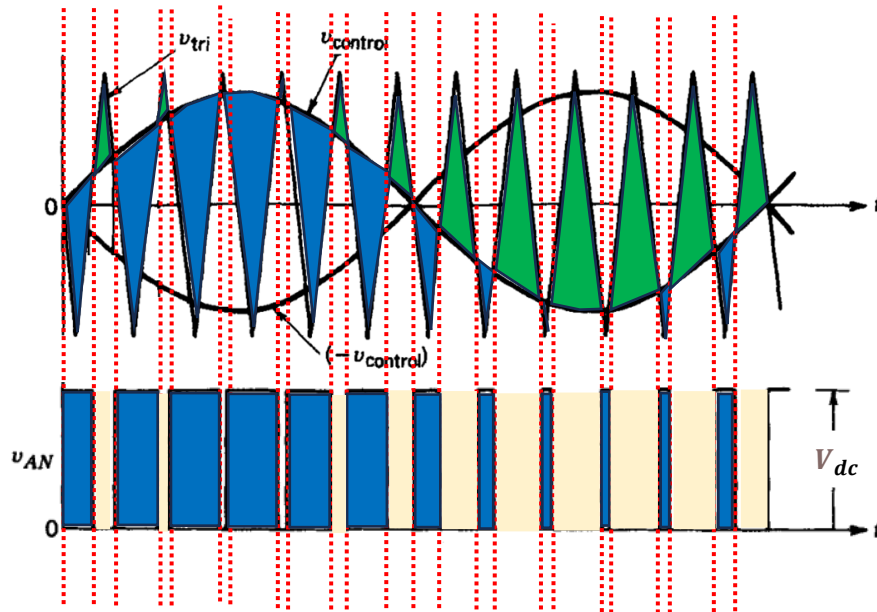
Operation

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 - Is a type of switching technique.
 - Use two signals called
 - v_{tri} (Triangular Signal)
 - Has frequency f_s (called switching frequency)
 - $v_{control}$ & $-v_{control}$ (Control Signal)
 - Has frequency f_1 (desired fundamental frequency)



Operation

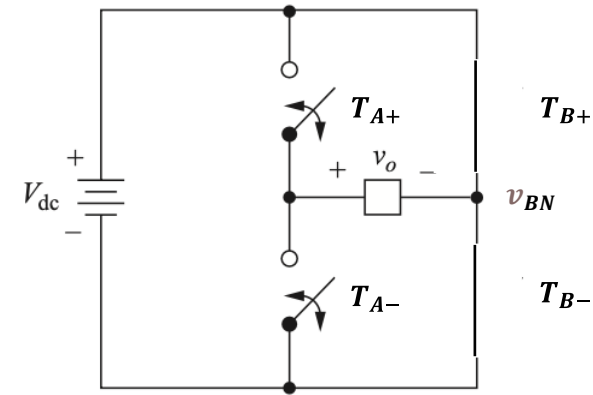
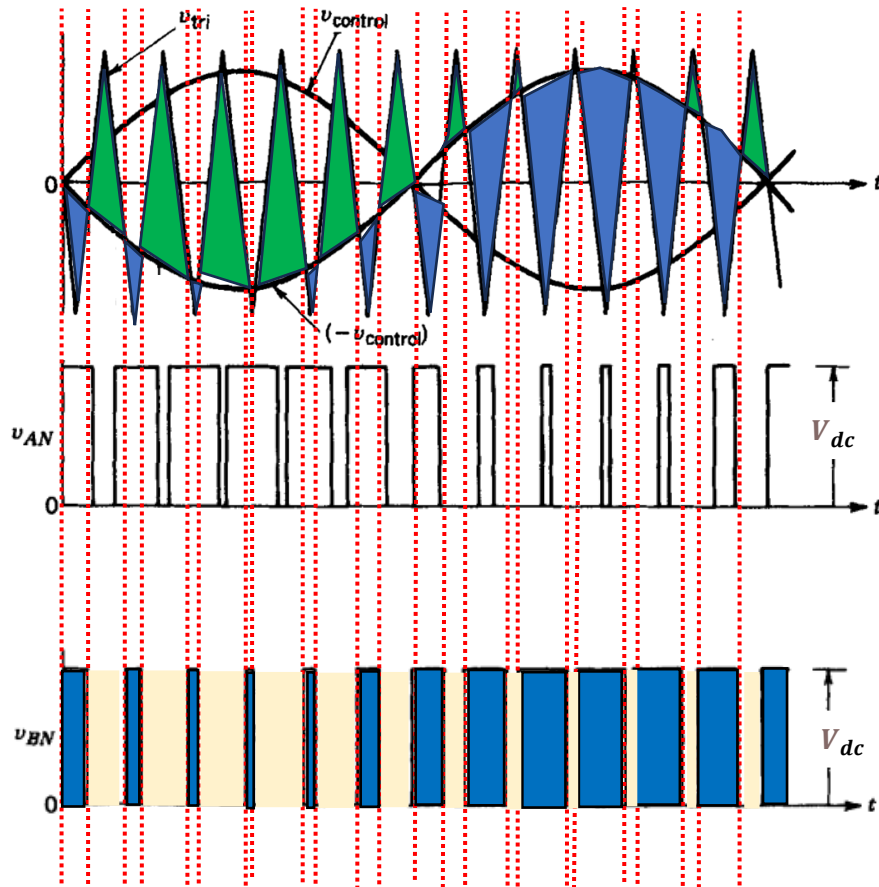
- Pulse Width Modulation (PWM) Switching - Unipolar
 - v_{tri} (Triangular Signal) is compared with $v_{control}$ & $-v_{control}$
 - Based on comparison result different switches are made closed (ON) and open (OFF)



$v_{control} > v_{tri}$:	T_{A+} on	and	$v_{AN} = V_{dc}$
$v_{control} < v_{tri}$:	T_{A-} on	and	$v_{AN} = 0$

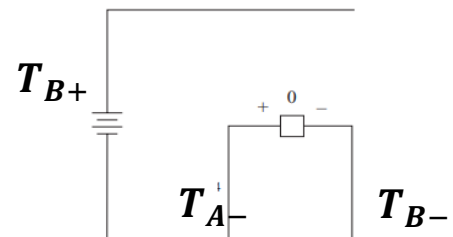
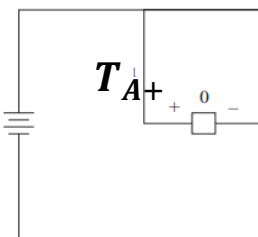
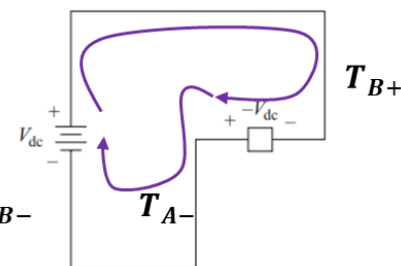
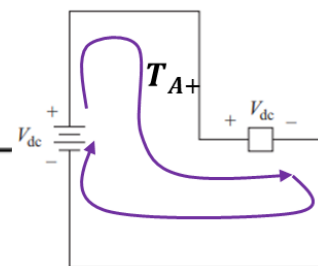
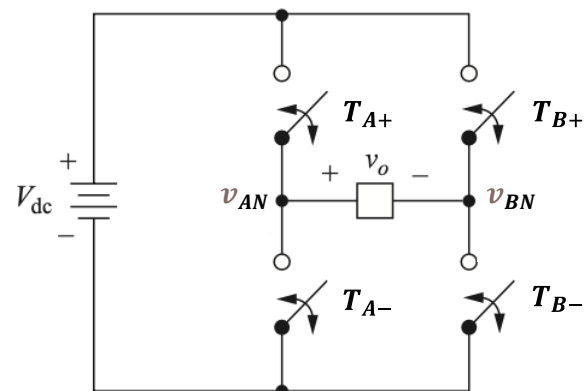
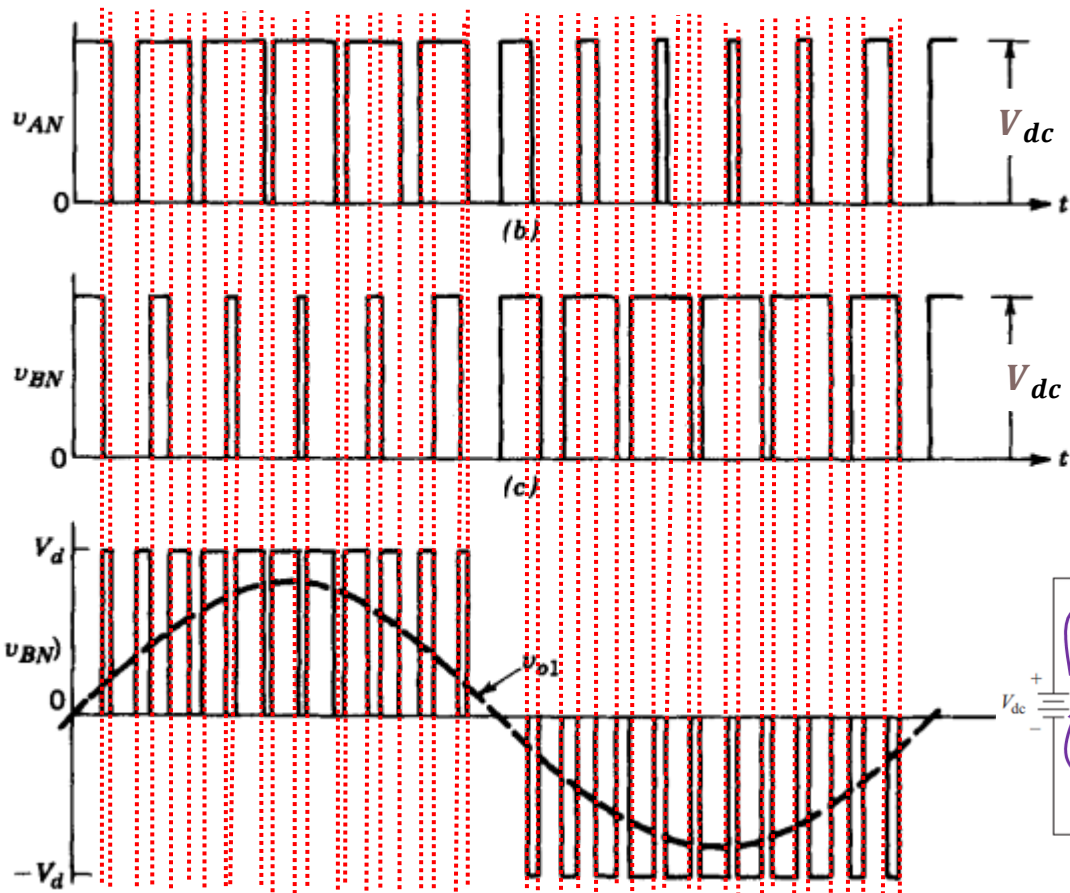
Operation

- Pulse Width Modulation (PWM) Switching - Unipolar
 - v_{tri} (Triangular Signal) is compared with $v_{control}$ & $-v_{control}$
 - Based on comparison result different switches are made closed (ON) and open (OFF)



$(-v_{control}) > v_{tri}$	T_{B+} on	and	$v_{BN} = V_{dc}$
$(-v_{control}) < v_{tri}$	T_{B-} on	and	$v_{BN} = 0$

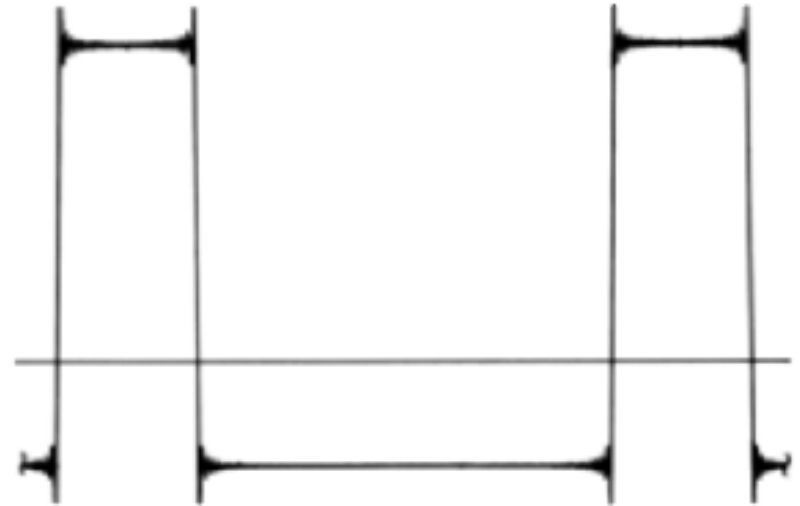
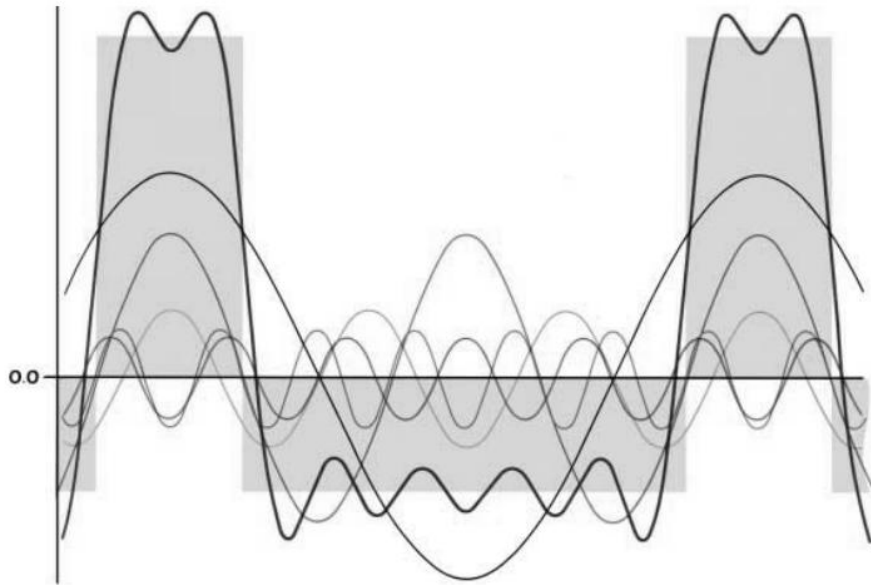
Operation



1. T_{A+}, T_{B-} on: $v_{AN} = V_{dc}$ $v_{BN} = 0$; $v_o = V_{dc}$
2. T_{A-}, T_{B+} on: $v_{AN} = 0$, $v_{BN} = V_{dc}$; $v_o = -V_{dc}$
3. T_{A+}, T_{B+} on: $v_{AN} = V_{dc}$ $v_{BN} = V_{dc}$ $v_o = 0$
4. T_{A-}, T_{B-} on: $v_{AN} = 0$, $v_{BN} = 0$; $v_o = 0$

Harmonics

- What are harmonics?
 - Pure Sinusoid is required.
 - The output is different both in square-wave and PWM switching
 - There is need to analyze the **presence of other sinusoids** having **frequencies multiple of desired sinusoid**, *called harmonics*, available in output.

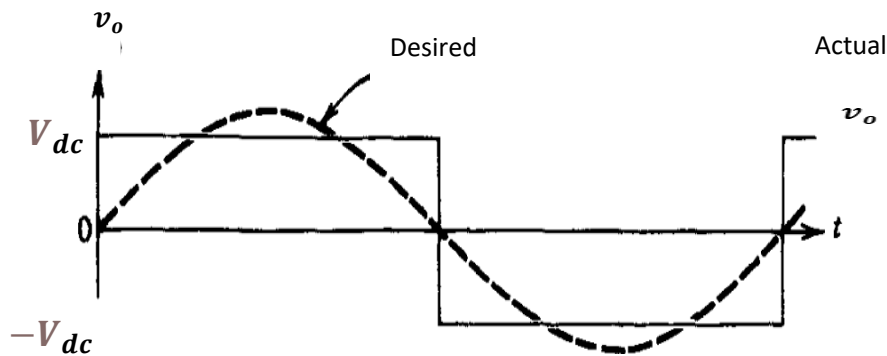
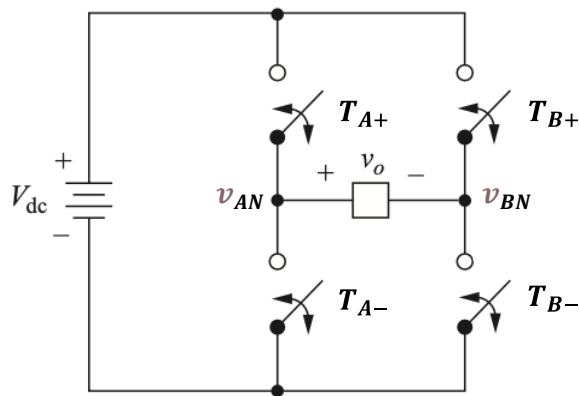


Rectangular waveshape produced by combining 100 harmonics. Note reduction in ripple.

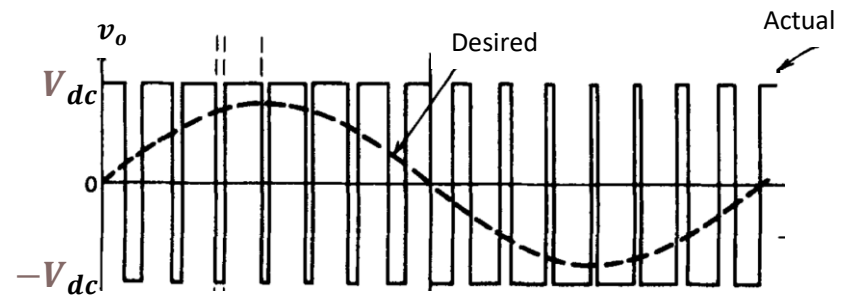
Harmonics

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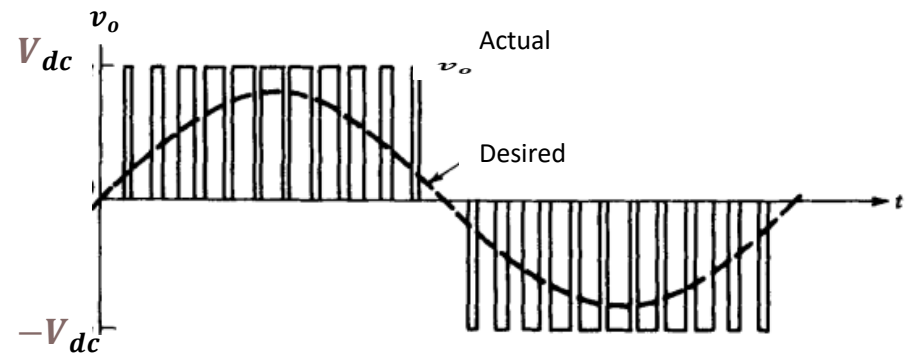
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Output (Square Wave Switching)



Output (PWM Switching - Bipolar)

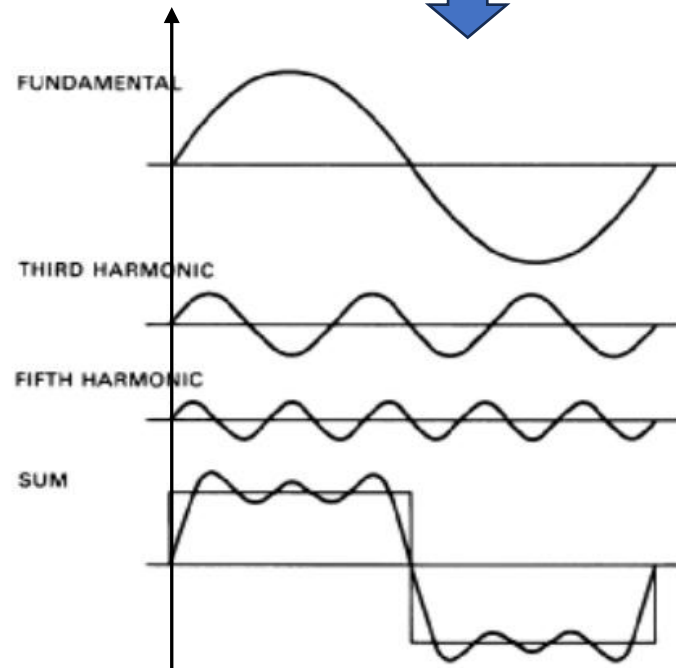
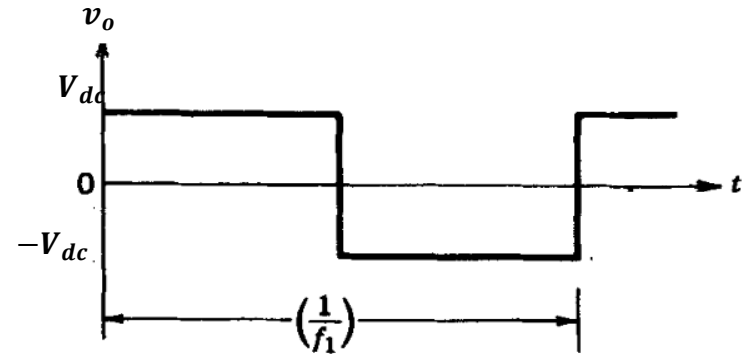
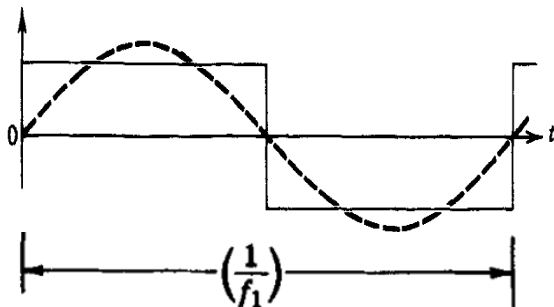
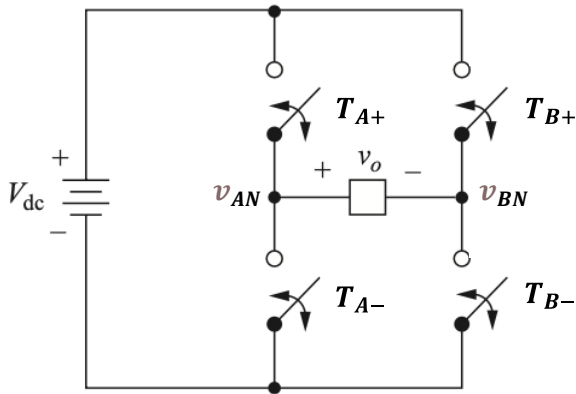


Output (PWM Switching - Unipolar)

Harmonics

- Harmonics in Square Wave Switching

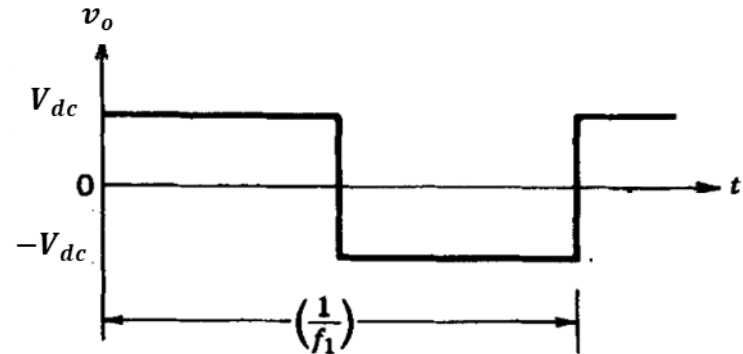
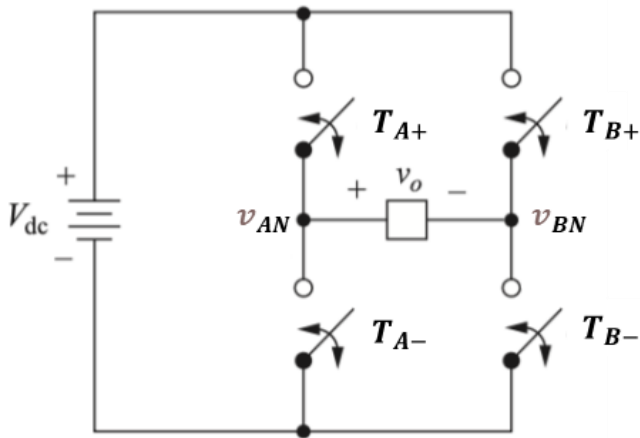
- There is need to analyze the **presence of other sinusoids** having **frequencies multiple of desired sinusoid, called harmonics**, available in output.



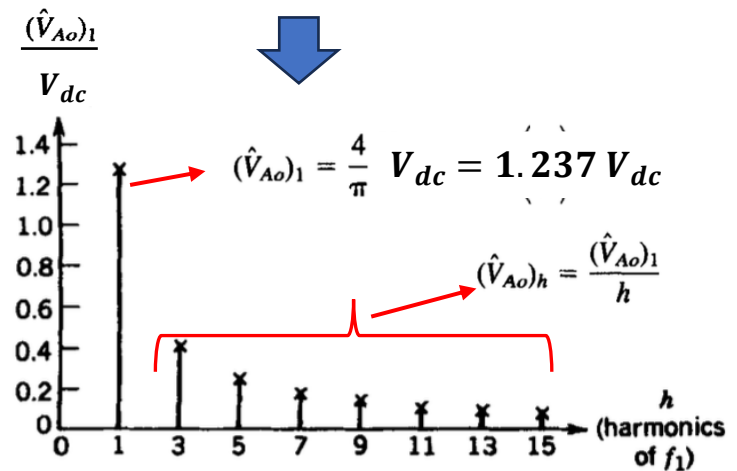
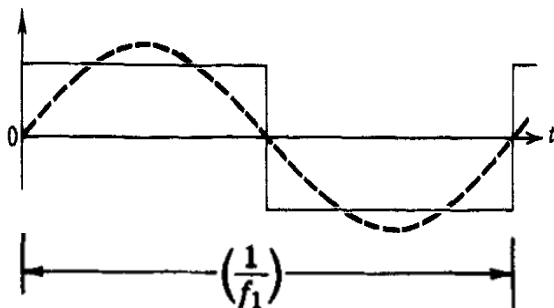
Harmonics

- Harmonics in Square Wave Switching

- There is need to analyze the **presence of other sinusoids** having **frequencies multiple of desired sinusoid, called harmonics**, available in output.



Fourier Series Analysis



Harmonics

- Harmonics in PWM Switching

- To analyze "*harmonics*" for PWM switching we need to understand two new terms:
 - **Amplitude Modulation Ratio** and **Frequency Modulation Ratio**.

$$m_a = \frac{\hat{V}_{\text{control}}}{\hat{V}_{\text{tri}}}$$



Amplitude Modulation Ratio

$$\hat{V}_{\text{control}}$$

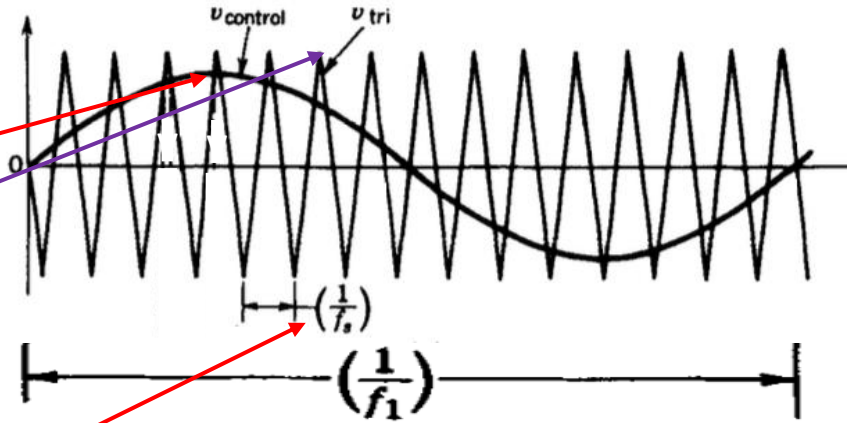


Peak Value of Control Signal

$$\hat{V}_{\text{tri}}$$



Peak Value of Triangular Signal



$$m_f = \frac{f_s}{f_1}$$



Frequency Modulation Ratio

$$f_s$$



Frequency of triangular signal
also called switching frequency

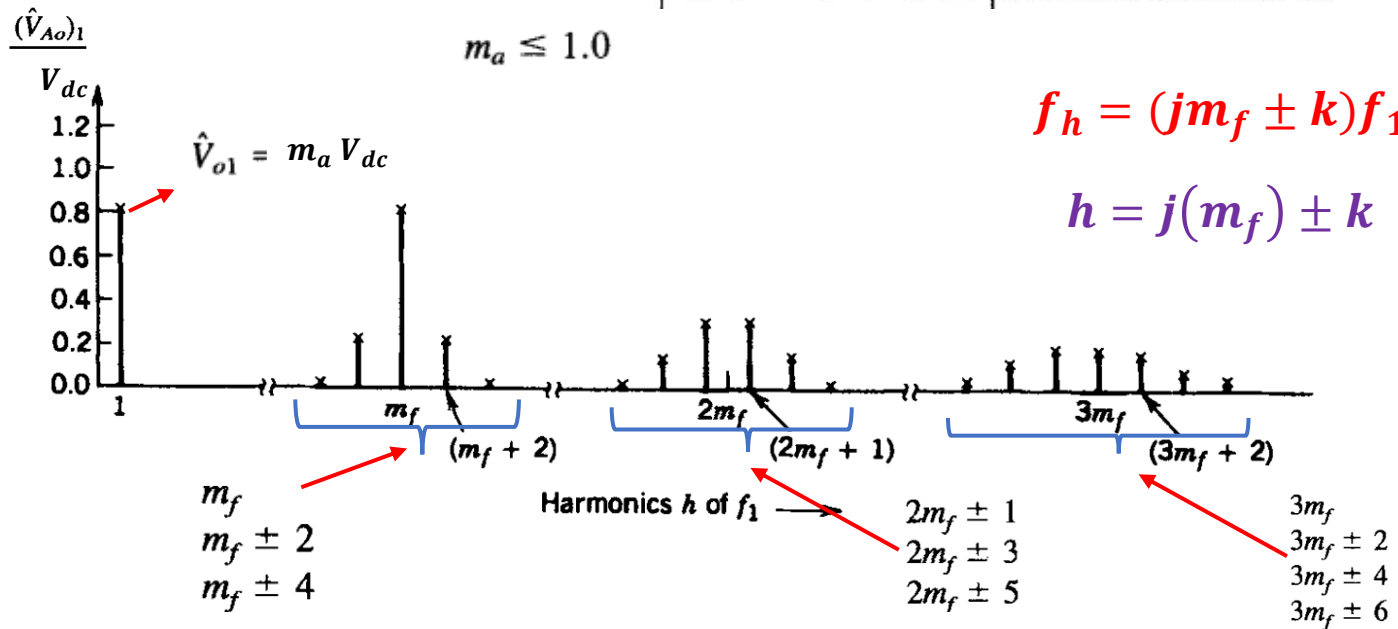
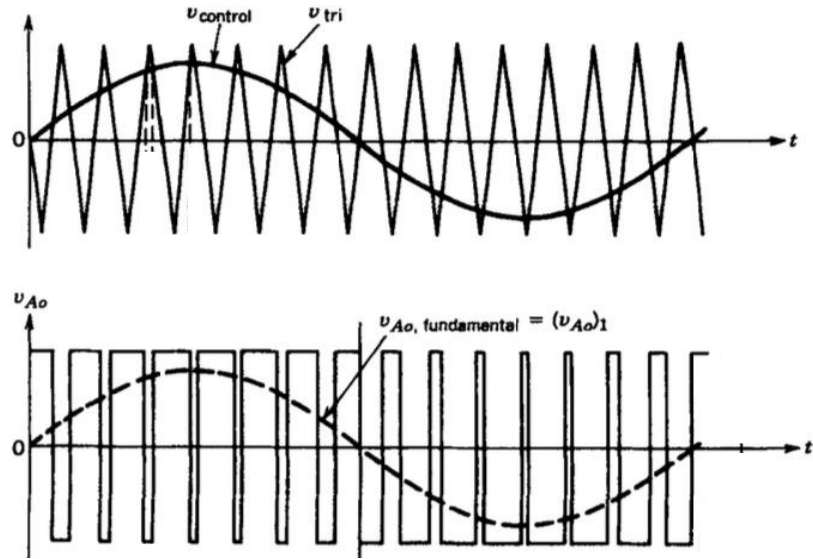
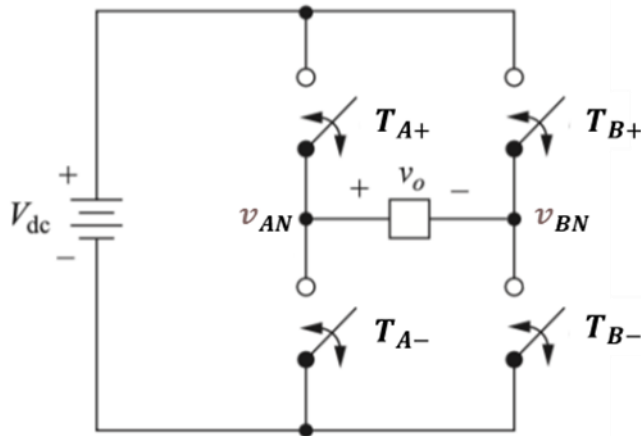
$$f_1$$



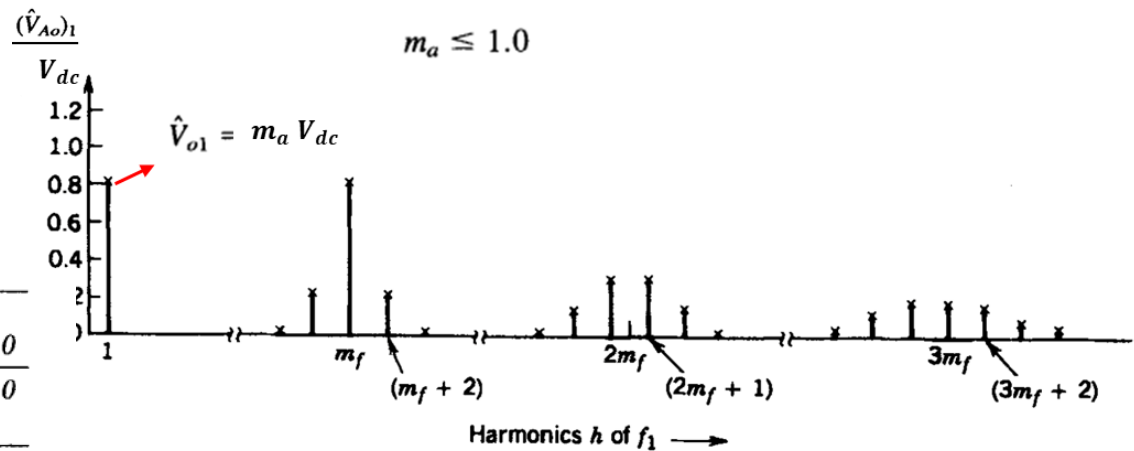
Frequency of control signal also
called desired fundamental frequency

Harmonics

- Harmonics in PWM Switching (Bipolar)



In the full-bridge converter circuit, $V_d = 300$ V, $m_a = 0.8$, $m_f = 39$, and the fundamental frequency is 47 Hz. Calculate the rms values of the fundamental-frequency voltage and some of the dominant harmonics in the output voltage v , if a PWM bipolar voltage-switching scheme is used.



$\begin{matrix} h \\ \backslash \\ m_a \end{matrix}$	0.2	0.4	0.6	0.8	1.0
1	0.2	0.4	0.6	0.8	1.0
Fundamental					
m_f	1.242	1.15	1.006	0.818	0.601
$m_f \pm 2$	0.016	0.061	0.131	0.220	0.318
$m_f \pm 4$					0.018
$2m_f \pm 1$	0.190	0.326	0.370	0.314	0.181
$2m_f \pm 3$		0.024	0.071	0.139	0.212
$2m_f \pm 5$				0.013	0.033
$3m_f$	0.335	0.123	0.083	0.171	0.113
$3m_f \pm 2$	0.044	0.139	0.203	0.176	0.062
$3m_f \pm 4$		0.012	0.047	0.104	0.157
$3m_f \pm 6$				0.016	0.044
$4m_f \pm 1$	0.163	0.157	0.008	0.105	0.068
$4m_f \pm 3$	0.012	0.070	0.132	0.115	0.009
$4m_f \pm 5$			0.034	0.084	0.119
$4m_f \pm 7$				0.017	0.050

$$V_{rms} = \frac{V_{peak}}{\sqrt{2}} = \frac{300}{\sqrt{2}} = 212.13$$

$$V_{o1} = V_{rms} \times m_a = 212.13 \times 0.8 = 169.7 \text{ V @ 47 Hz}$$

$$(V_o)_{37} = 212.13 \times 0.22 = 46.67 \text{ V at 1739 Hz}$$

$$(V_o)_{39} = 212.13 \times 0.818 = 173.52 \text{ V at 1833 Hz}$$

$$(V_o)_{41} = 212.13 \times 0.22 = 46.67 \text{ V at 1927 Hz}$$

$$(V_o)_{77} = 212.13 \times 0.314 = 66.60 \text{ V at 3619 Hz}$$

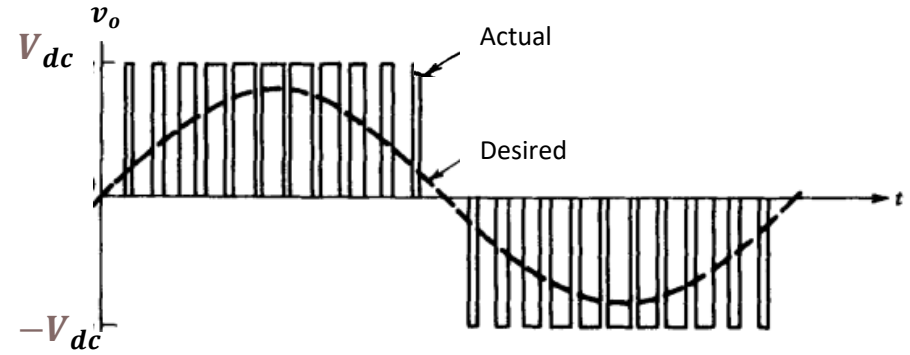
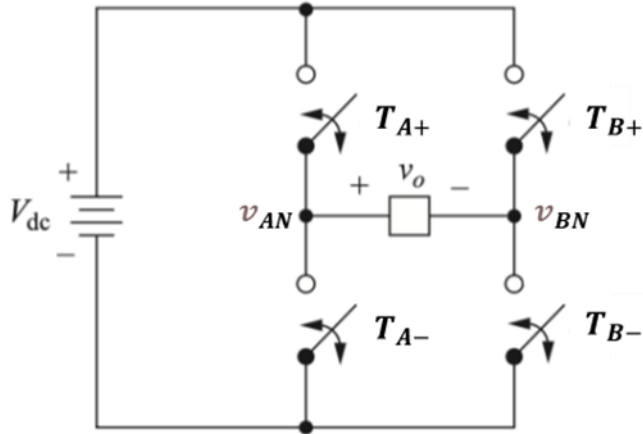
$$(V_o)_{79} = 212.13 \times 0.314 = 66.60 \text{ V at 3713 Hz}$$

etc

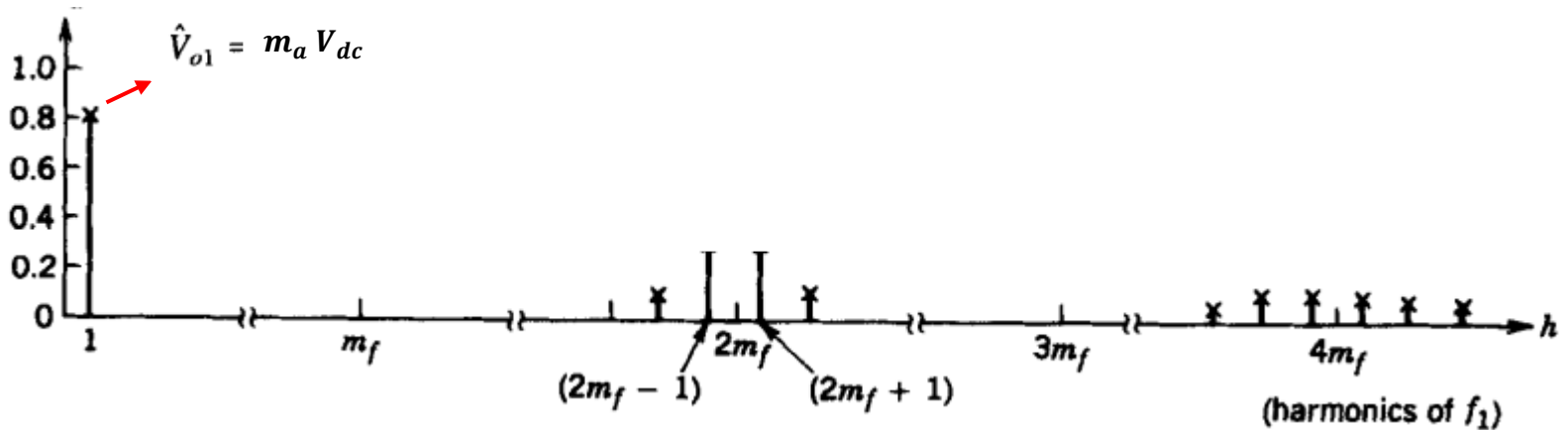
$\frac{(\hat{V}_{Ao})_h}{V_d}$ values at different values of m_a

Harmonics

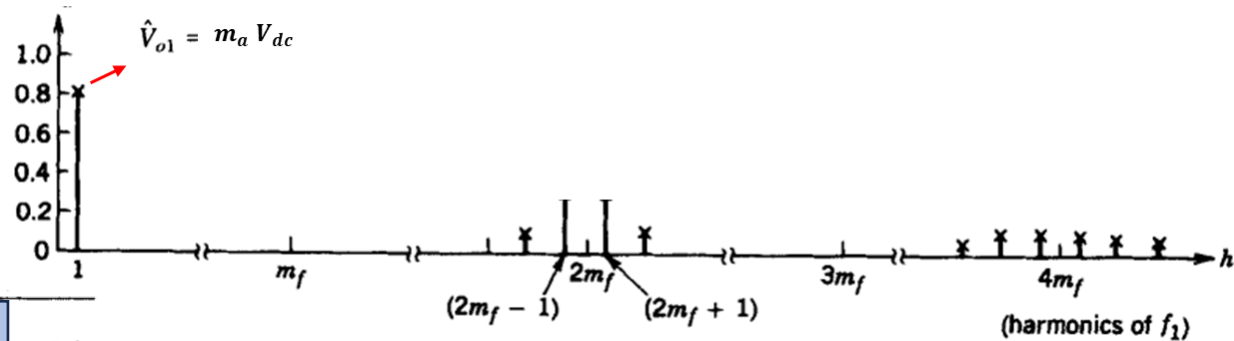
- Harmonics in PWM Switching (Unipolar)



Output (PWM Switching - Unipolar)



In the full-bridge converter circuit, $V_d = 300$ V, $m_a = 0.8$, $m_f = 38$, and the fundamental frequency is 47 Hz. Calculate the rms values of the fundamental-frequency voltage and some of the dominant harmonics in the output voltage v , if a PWM bipolar voltage-switching scheme is used.



$\begin{matrix} m_a \\ h \end{matrix}$	0.2	0.4	0.6	0.8	1.0
1	0.2	0.4	0.6	0.8	1.0
Fundamental					
m_f	1.242	1.15	1.006	0.818	0.601
$m_f \pm 2$	0.016	0.061	0.131	0.220	0.318
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$3m_f \pm 2$	0.044	0.139	0.203	0.176	0.062
$3m_f \pm 4$		0.012	0.047	0.104	0.157
$3m_f \pm 6$				0.016	0.044
$4m_f \pm 1$	0.163	0.157	0.008	0.105	0.068
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$4m_f \pm 5$			0.034	0.084	0.119
$4m_f \pm 7$				0.017	0.050

$$V_{rms} = \frac{V_{peak}}{\sqrt{2}} = \frac{300}{\sqrt{2}} = 212.13$$

$$V_{o1} = V_{rms} \times m_a = 212.13 \times 0.8 = 169.7 \text{ V @ 47 Hz}$$

$$\text{At } h = 2m_f - 1 = 75 \text{ or } 3525 \text{ Hz:}$$

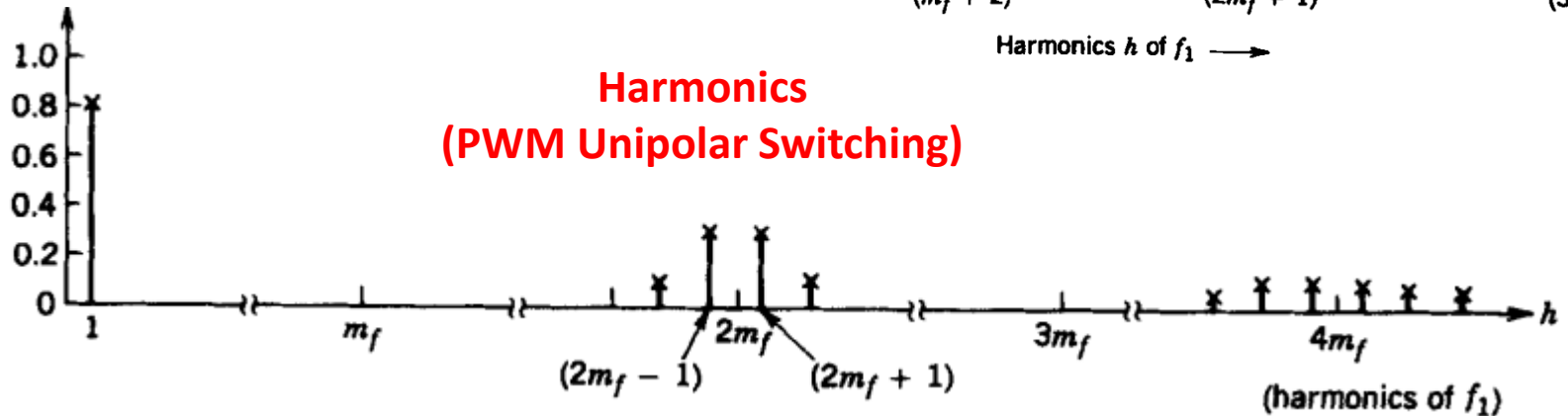
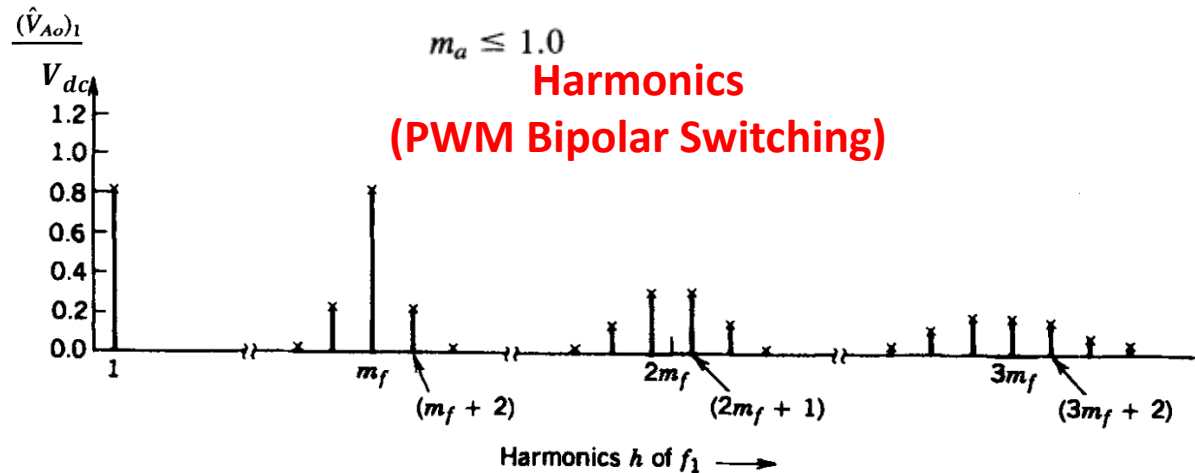
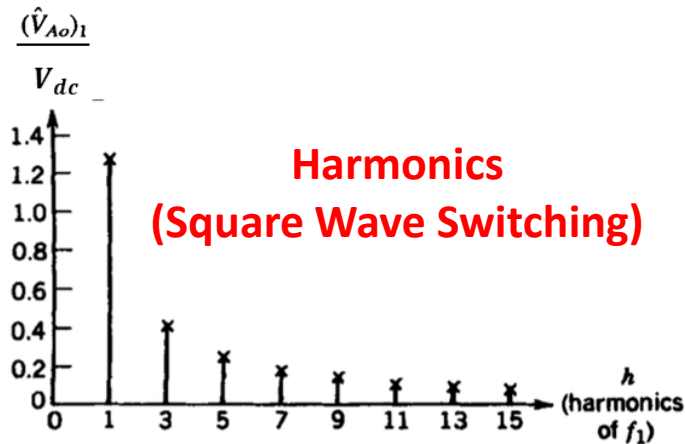
$$(V_o)_{75} = 0.314 \times 212.13 = 66.60 \text{ V}$$

$$\text{At } h = 2m_f + 1 = 77 \text{ or } 3619 \text{ Hz:}$$

$$(V_o)_{77} = 0.314 \times 212.13 = 66.60 \text{ V}$$

$\frac{(\hat{V}_{Ao})_h}{V_d}$ values at different values of m_a

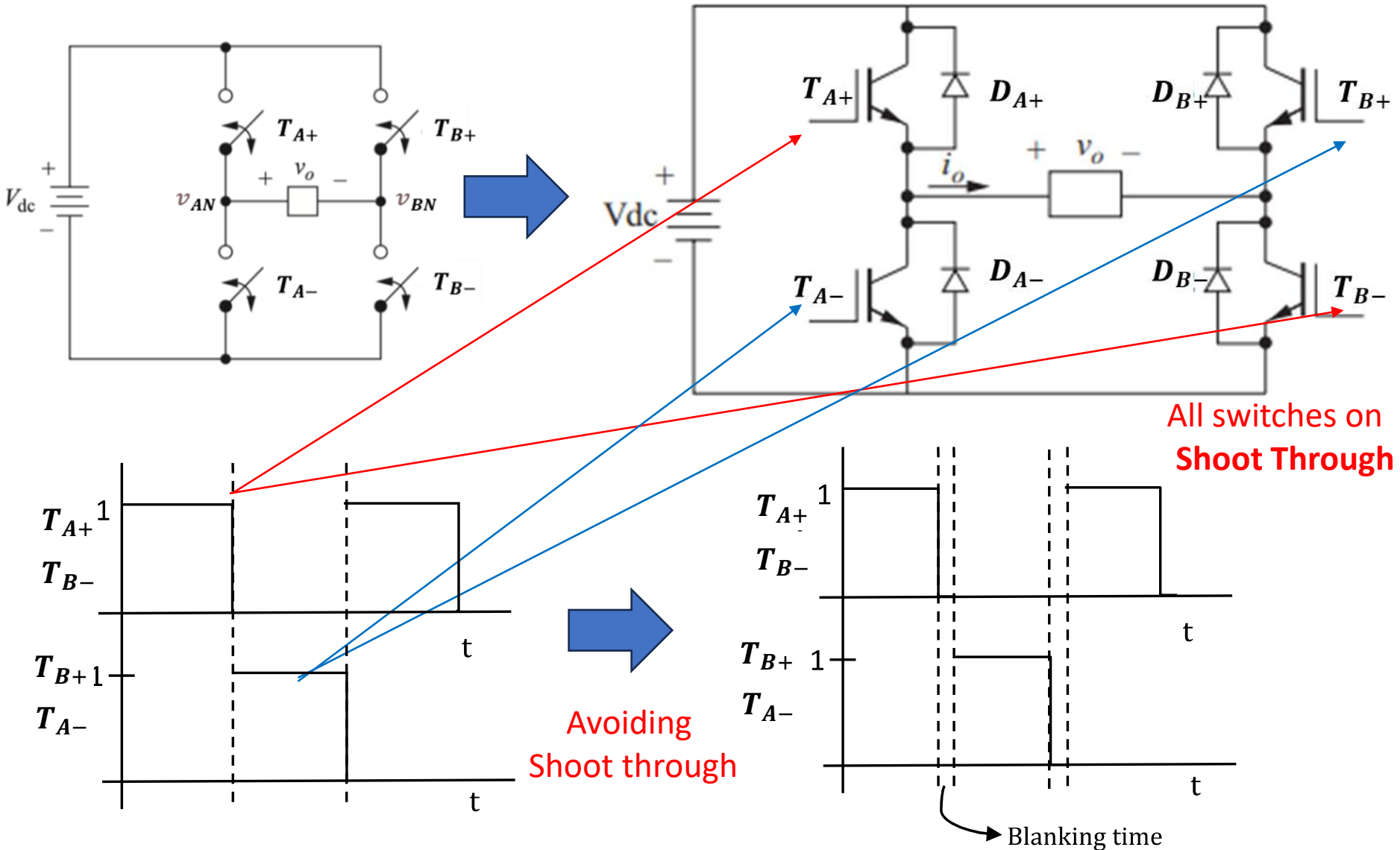
Harmonics - Comparing Harmonics



- The Unipolar scheme has the advantage of “effectively” doubling the switching frequency as far as the output harmonics are concerned, compared to the bipolar voltage-switching scheme.
- The advantage of “effectively” doubling the switching frequency appears in the harmonic spectrum of the output voltage waveform, where the lowest harmonics appear as sidebands of twice the switching frequency.**

Practical Full Bridge Inverter

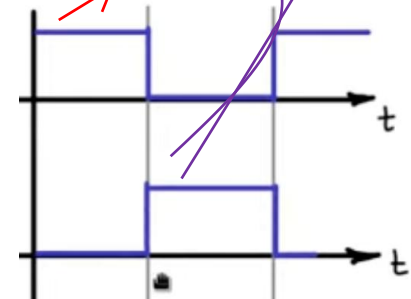
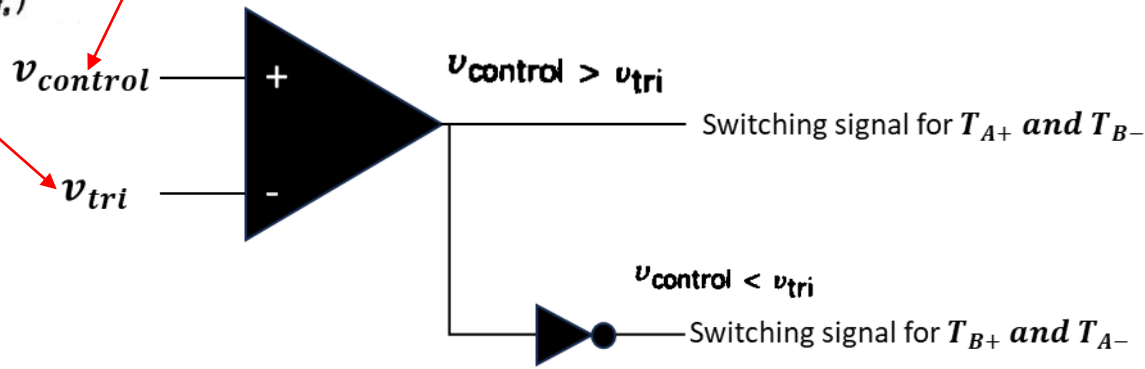
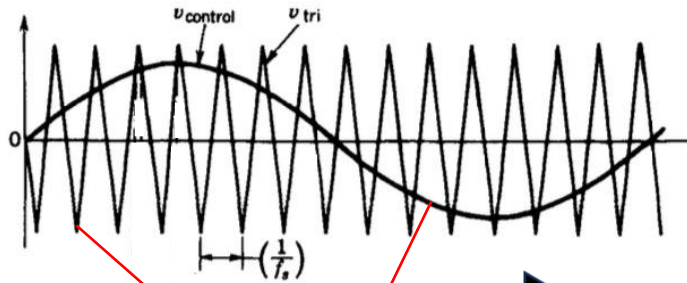
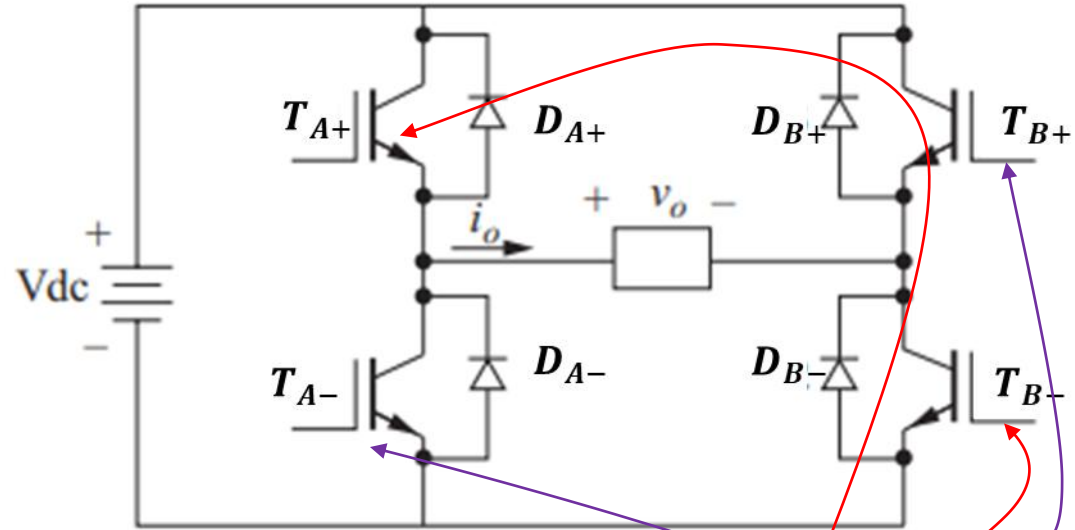
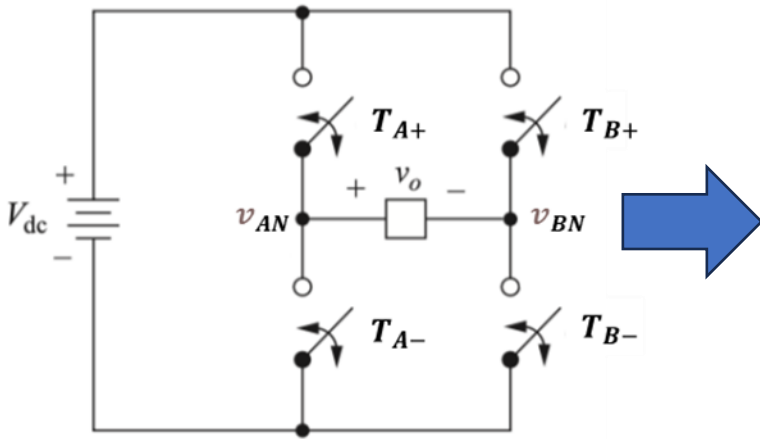
- Made through real components
 - Square Wave Switching



Practical Full Bridge Inverter

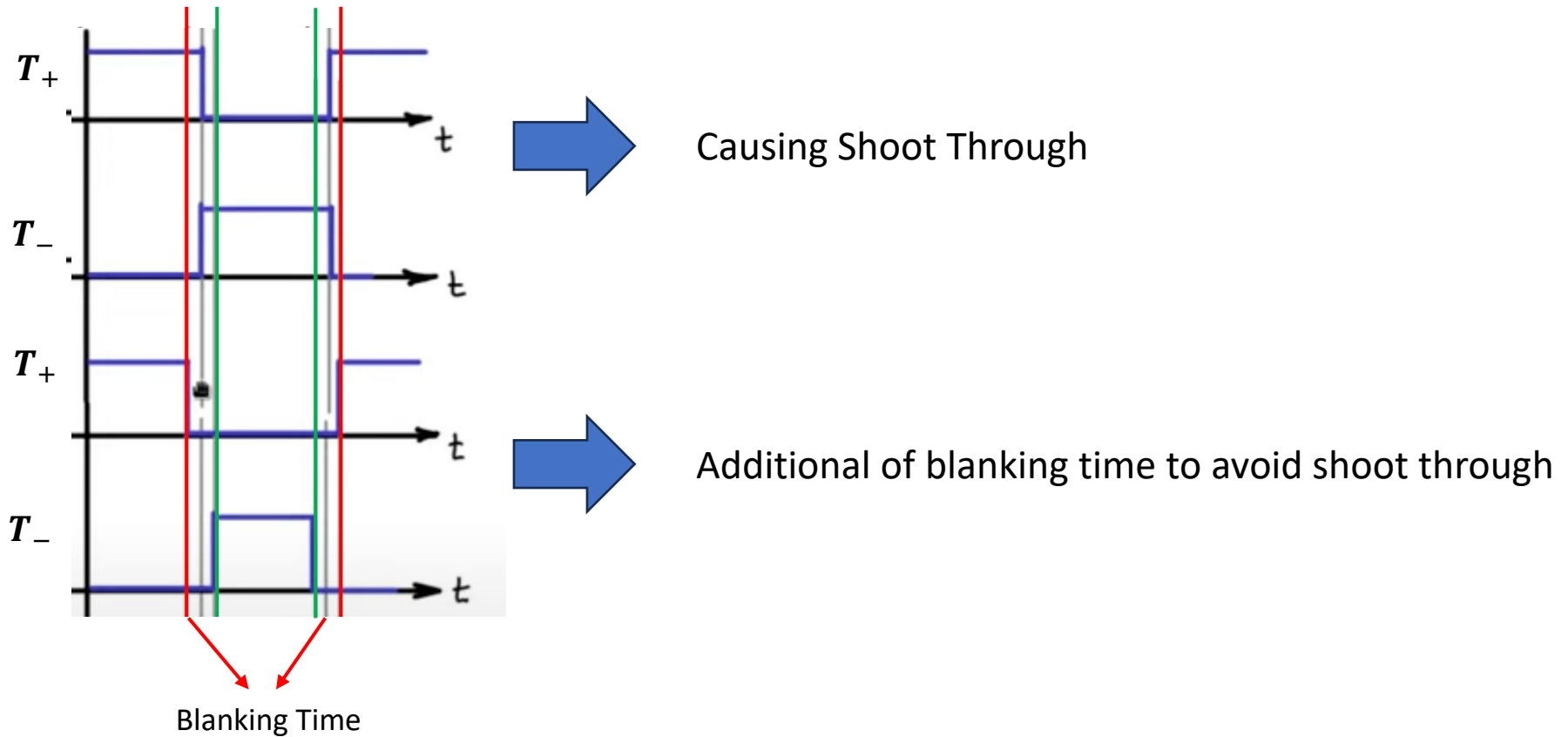
- Made through real components

- PWM Switching - **Bipolar**



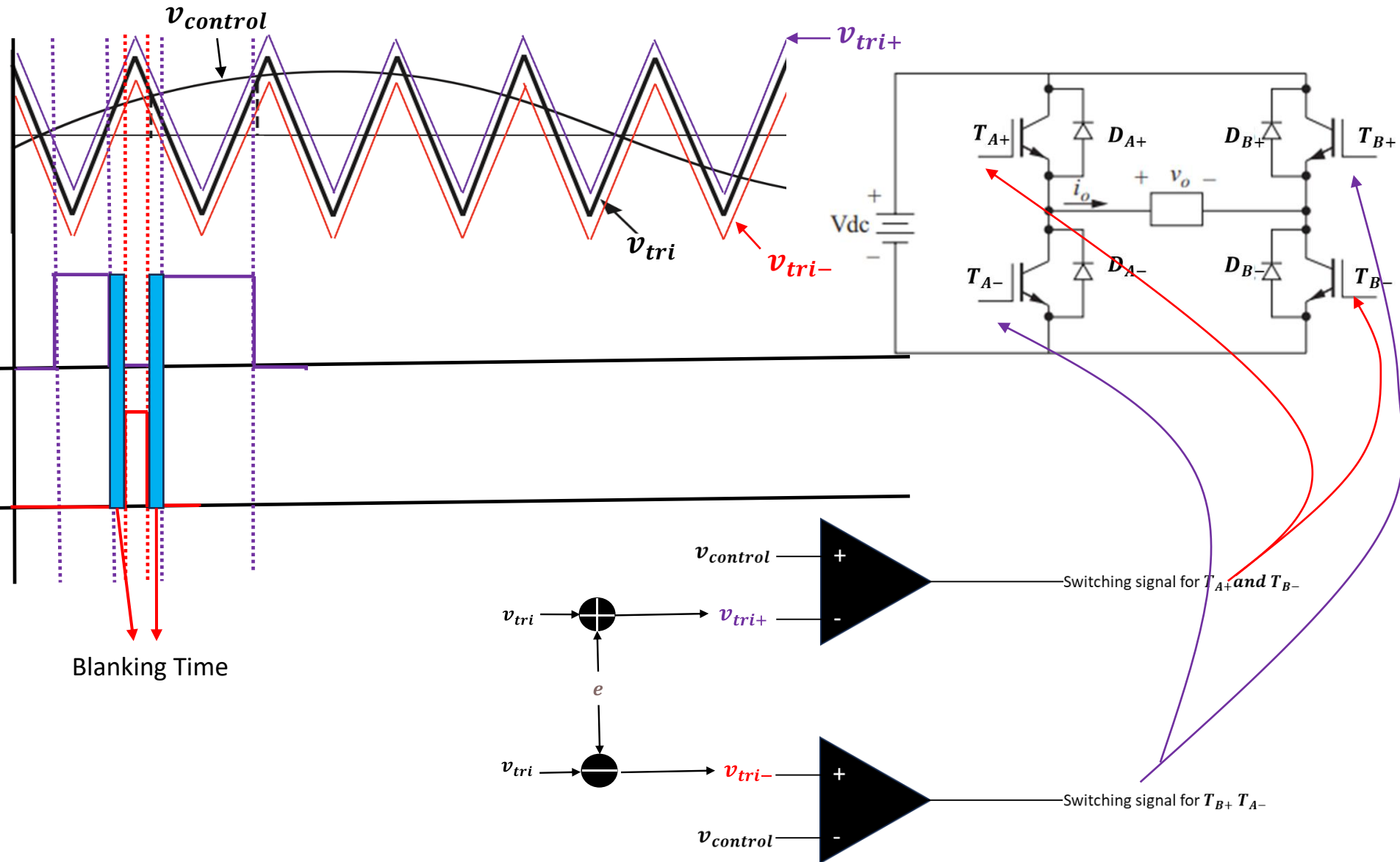
Practical Full Bridge Inverter

- Made through real components
 - Avoiding shoot through in PWM – **Bipolar Switching**



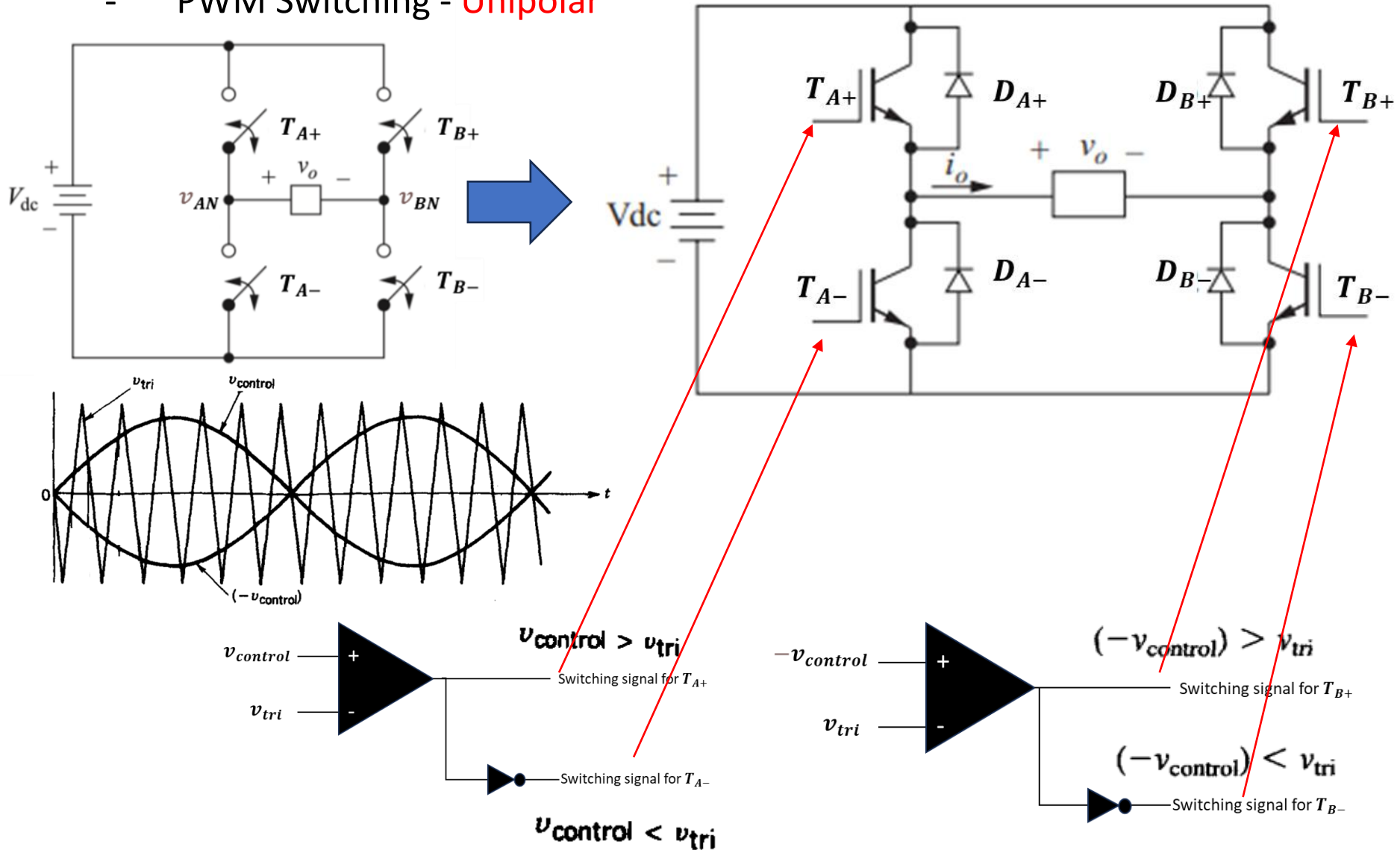
Avoiding Shoot through

- Avoiding shoot through in PWM – **Bipolar Switching**

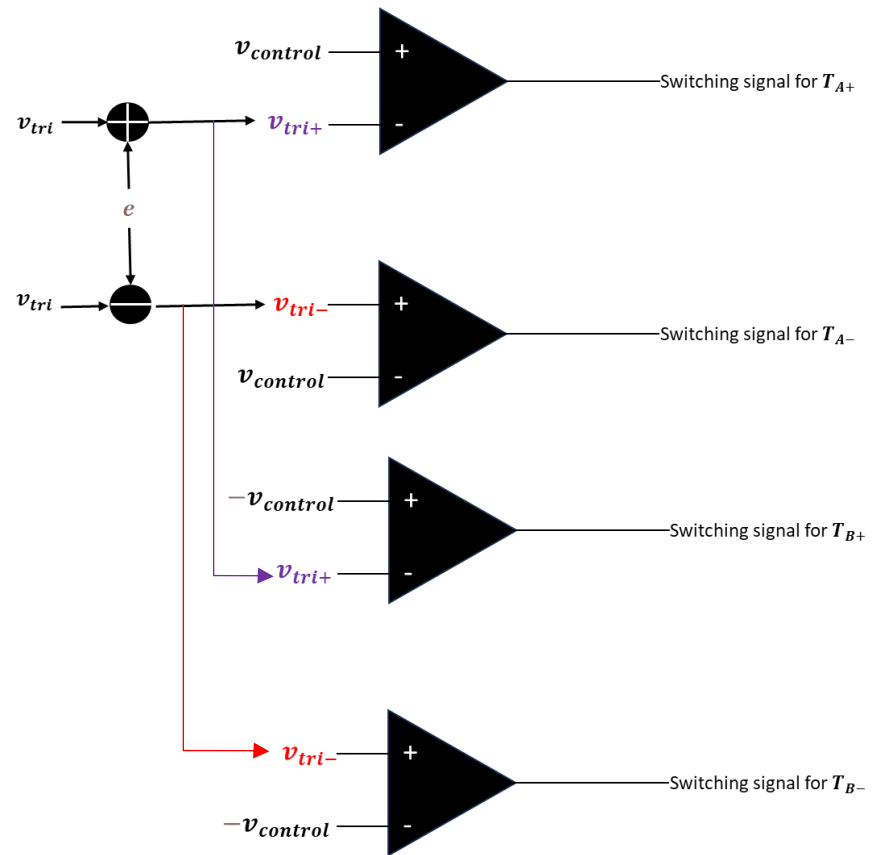
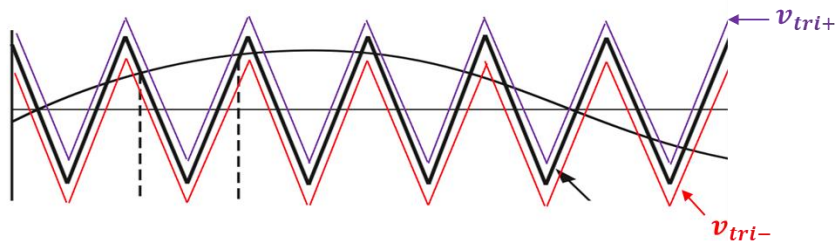


Practical Full Bridge Inverter

- Made through real components
 - PWM Switching - **Unipolar**



To avoid shoot through in
Unipolar two circuits made for
Bipolar shall be used



Comparison of different Schemes

	Half Bridge ~ Full Bridge
Output Power Square Wave Switching	With the same dc input voltage, the maximum output voltage of the half-bridge inverter is half that of the full-bridge inverter. Hence, in case of low power applications we can use the half-bridge inverters and save one-leg of full bridge inverter.

Comparison of different Schemes

	Square Wave Switching	PWM Switching
Harmonics in Half or Full Bridge Inverter	The abrupt switching in square waves generates a significant number of harmonics. Harmonics are multiples of the fundamental frequency (the desired AC output frequency) that distort the waveform and can cause problems for connected equipment.	PWM uses a more sophisticated approach. It generates a pulse train with a constant frequency (carrier frequency) but varies the width of each pulse (duty cycle). By carefully controlling the duty cycle, PWM can effectively synthesize a sinusoidal output voltage. This approach "spreads out" the harmonics to higher frequencies, making them easier to filter.
	To mitigate these harmonics and achieve a cleaner output waveform, square wave inverters need bulky and expensive filtering components.	As a result of this harmonic shaping, PWM inverters require smaller and more efficient filters to achieve the same level of output quality compared to square wave inverters.

Comparison of different Schemes

	Bipolar ~ Unipolar
Harmonics PWM Switching	In unipolar scheme, the advantage of “effectively” doubling the switching frequency appears in the harmonic spectrum of the output voltage waveform, where the lowest harmonics appear as sidebands of twice the switching frequency.

Summary

- Full Bridge DC-AC Converter studied with following switching option:
 - Square Wave Switching
 - PWM Bipolar Switching
 - PWM Unipolar switching
- following aspects were studied for each switching:
 - Ideal Circuit Diagram
 - Operation
 - Output Voltage (Waveform)
 - Harmonics
 - Practical Circuit Diagram
 - Avoiding shoot through
- A comparison is made in different single-phase inverters.